

TA Proposal: SQL & NoSQL Integrated Assignments

Objective

To enhance students' understanding of Database Systems by integrating both SQL and NoSQL based assignments that reflect real-world system design decisions, performance considerations, and scalability trade-offs.

Why This Approach Is Different

- Moves beyond textbook SQL queries to system-level thinking
- Introduces modern database usage seen in industry
- Encourages comparison-based learning (SQL vs NoSQL)
- Reduces copy-based assignments through design-oriented tasks

Assignment Structure

Each assignment is divided into two parallel tracks using the same problem domain:

- **Part A – SQL Implementation:** Schema design, normalization, joins, indexing, and query optimization
- **Part B – NoSQL Implementation:** Document modeling, embedding vs referencing, and query patterns

Sample Assignment Example

- Design a database for a Student Expense Tracker system
- Implement relational schema with proper normalization in MySQL/PostgreSQL
- Implement the same system using MongoDB with document modeling
- Compare performance, scalability, and flexibility of both approaches

Performance & Optimization Component

- Use of indexes and EXPLAIN plans in SQL
- Query pattern optimization in MongoDB
- Identification of slow queries and redesigning schemas

Incorporating My Experience

- Experience building full-stack systems using SQL and MongoDB
- Understanding of real-world database failures and performance issues
- Ability to guide students on choosing the right database for a problem

Expected Learning Outcomes

- Strong conceptual understanding of relational and non-relational databases
- Ability to design efficient schemas for different workloads
- Practical knowledge aligned with industry practices

This approach aims to prepare students not only to pass examinations, but to design scalable, efficient, and maintainable database systems in real-world environments.